// **1**) input data for 2 students.

```
#include<stdio.h>
struct student{
	char name[50];
	int age;
	float total_mark;
};
int main(){
	struct student a[2];
	int i;
	for (i=0;i<2;i++){
		printf("Enter the name of the student: ");
		scanf(" %[^\n]s",a[i].name);
		printf("Enter the age of the student: ");
		scanf(" %d",&a[i].age);
		printf("Enter the total mark of the student: ");
		scanf(" %f",&a[i].total_mark);
		printf("\n");
	}
	for (i=0;i<2;i++){
		printf("The name of the student : %s\n",a[i].name);
		printf("The age of %s : %d\n",a[i].name,a[i].age);
		printf("The total mark of %s : %.2f\n\n",a[i].name,a[i].total_mark);

	}
	return 0;
}
```

```
Enter the name of the student: Joshan Koirala
Enter the age of the student: 19
Enter the total mark of the student: 500

Enter the name of the student: Sagar Koirala
Enter the age of the student: 20
Enter the total mark of the student: 490

The name of the student : Joshan Koirala
The age of Joshan Koirala : 19
The total mark of Joshan Koirala : 500.00

The name of the student : Sagar Koirala
The age of Sagar Koirala : 20
The total mark of Sagar Koirala : 490.00


--------------------------------
Process exited after 35.35 seconds with return value 0
```

*//**2)** sum of input times.*

```
#include<stdio.h>
struct Time{
        int hours;
        int minutes;
        int seconds;
};
int main() {
   struct Time t1, t2, result;
   struct Time sum = {0,0,0};
   printf("Enter hours for the first time: ");
   scanf("%d", &t1.hours);
   printf("Enter minutes for the first time: ");
   scanf("%d", &t1.minutes);
   printf("Enter seconds for the first time: ");
   scanf("%d", &t1.seconds);
   printf("\n");
   printf("Enter hours for the second time: ");
   scanf("%d", &t2.hours);
   printf("Enter minutes for the second time: ");
   scanf("%d", &t2.minutes);
   printf("Enter seconds for the second time: ");
   scanf("%d", &t2.seconds);
   printf("\n");
   // operation
    sum.seconds = t1.seconds + t2.seconds;
   if (sum.seconds >= 60) {
       sum.minutes += sum.seconds/60;
       sum.seconds =  sum.seconds%60;
   }
   sum.minutes += t1.minutes + t2.minutes;
   if (sum.minutes >= 60){
      sum.hours += sum.minutes/60;
      sum.minutes = sum.minutes%60;
  }
   sum.hours += t1.hours + t2.hours;
   result = sum;
```

```c
printf ("Sum of the two times:%d hours %d minutes and    %d
        seconds\n",result.hours,result.minutes,result.seconds);

  return 0;

}
```

```
E:\C Programming\Struct_Tim

Enter hours for the first time: 6
Enter minutes for the first time: 22
Enter seconds for the first time: 35

Enter hours for the second time: 11
Enter minutes for the second time: 54
Enter seconds for the second time: 43

Sum of the two times:18 hours 17 minutes and 18 seconds

--------------------------------
Process exited after 22.24 seconds with return value 0
Press any key to continue . . .
```

*// **3**) Book with price between 2000 to 3000 with input of 5 book.*

```
#include<stdio.h>
struct Book{
        char title[50];
        char author[50];
        int price;
}b[5];
int main(){
        int i;
        for(i=0;i<5;i++){
                printf("Enter the title of the book: ");
                scanf(" %[^\n]s",b[i].title);
                getchar();
                printf("Enter the author of the book: ");
                scanf(" %[^\n]s",b[i].author);
                getchar();
                printf("Enter the price of the book: ");
                scanf("%d",&b[i].price);
                getchar();
                printf("\n");
        }
        printf("Book with price between 2000 and 3000 are:\n");
        int found=0;
        for(i=0;i<5;i++){
         if (b[i].price > 2000 && b[i].price < 3000){
            printf("Title: %s, Author: %s, Price: %d\n", b[i].title, b[i].author, b[i].price);
         }
         found=1;
  }

  if (!found) {
    printf("No books found in this price range.\n");
  }
        return 0;
}
```

E:\C Programming\StructureF

```
Enter the title of the book: Behal Zindagi
Enter the author of the book: Joshan Koirala
Enter the price of the book: 2980

Enter the title of the book: Physics
Enter the author of the book: Prajwal Bhandari
Enter the price of the book: 3000

Enter the title of the book:  Programming
Enter the author of the book: Ribshav Poudyal
Enter the price of the book: 2870

Enter the title of the book: Digital Logics
Enter the author of the book: Nawaraj Joshi
Enter the price of the book: 10000

Enter the title of the book: Mathematics
Enter the author of the book: Mohit Guragai
Enter the price of the book: 4000

Book with price between 2000 and 3000 are:
Title: Behal Zindagi, Author: Joshan Koirala, Price: 2980
Title: Programming, Author: Ribshav Poudyal, Price: 2870

--------------------------------
Process exited after 346.2 seconds with return value 0
Press any key to continue . . .
```

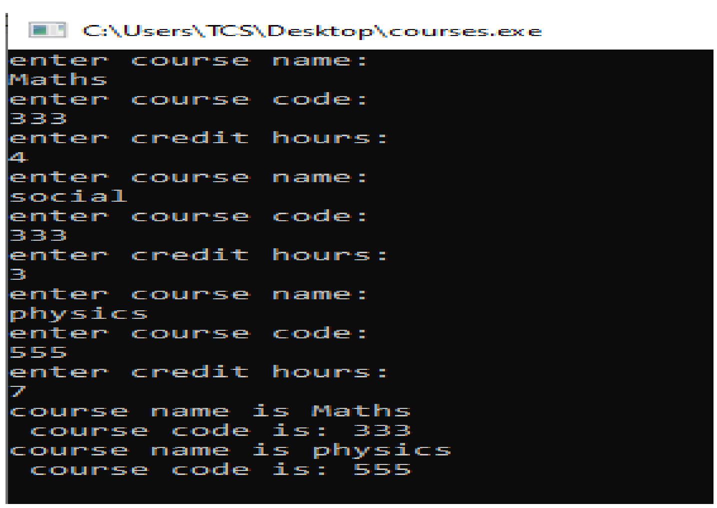
```c
//Program 4 of Structure:
#include<stdio.h>
struct complex{
	int real;
	int img;
} a[2];
int main(){
	int r,i;
	printf("enter the value for 1st complex number\n");
	printf("enter real part:\n");
	scanf("%d",&a[0].real);
	printf("enter imaginary part:\n");
	scanf("%d",&a[0].img);
	printf("1st complex number is: %d + i%d\n",a[0].real,a[0].img);
	printf("enter the value for 2nd complex number\n");
	printf("enter real part:\n");
	scanf("%d",&a[1].real);
	printf("enter imaginary part:\n");
	scanf("%d",&a[1].img);
	printf("2nd complex number is: %d + i%d\n",a[1].real,a[1].img);
	r=a[0].real+a[1].real;
	i=a[0].img+a[1].img;
	printf("the sum of two complex number is: %d + i%d\n",r,i);
	return 0;
}
```

Output:

```
D:\Lab Report c From Structure\file in c\complex.exe
enter the value for 1st complex number
enter real part:
3
enter imaginary part:
4
1st complex number is: 3 + i4
enter the value for 2nd complex number
enter real part:
2
enter imaginary part:
3
2nd complex number is: 2 + i3
the sum of two complex number is: 5 + i7
```

```
//program 5 of structure;
#include<stdio.h>
#include<string.h>
struct courses{
char c_name[30];
int c_code;
int credit_hours;
} a[3];
int main(){
	int i;
	for(i=0;i<3;i++){
		printf("enter course name:\n");
		scanf("%[^\n]s",a[i].c_name);
		getchar();
		printf("enter course code:\n");
		scanf("%d",&a[i].c_code);
			getchar();
		printf("enter credit hours:\n");
		scanf("%d",&a[i].credit_hours);
			getchar();
	}
	for(i=0;i<3;i++){
		if(a[i].credit_hours>3){
			printf("course name is %s \n course code is:
%d\n",a[i].c_name,a[i].c_code);
		}
	}
	return 0;
}
```

Output:



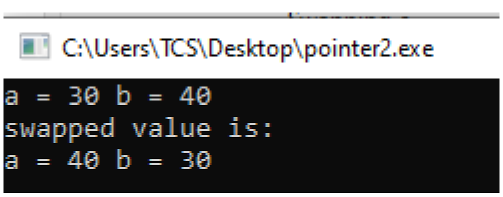
```
//first pointer program
#include<stdio.h>
int main(){
	int *p,*q,*r;
	int a =90,b=50;
	p=&a;
	q=&b;
	int c = *p+*q;
	r = p-q;
	printf("the value of variable a is :%d\n",*p);
	printf("address is : %d\n",p);
	printf("sum is %d \n",c);
	printf("the diff is :%d\n",r);
	if(p>q){
		printf("trueee\n");
	}
	else{
		printf("not\n");
	}
	return 0;
}
```

Output:

```
C:\Users\TCS\Desktop\pointer1.exe
the value of variable a is :90
address is : 6487548
sum is 140
the diff is :1
trueee
```

```c
//second program.
#include<stdio.h>
int main(){
	int *p,*q;
	int a=30,b=40;
	printf("a = %d b = %d\n",a,b);
	int temp;
	p=&a, q = &b;
	temp = *p;
	*p=*q;
	*q=temp;
	printf("swapped value is: \n");
	printf("a = %d b = %d",a,b);
	return 0;
}
```

Output:



```c
//pointer and array:
#include<stdio.h>
int main(){
	int n;
	printf("how many elements do you wanna store:\n");
	scanf("%d",&n);
	int a[n];
	printf("enter elements for the array:\n");
	int i;
	for(i=0;i<n;i++){
		scanf("%d",a+i);
	}
	for(i=0;i<n;i++){
		printf("array[%d]=%d\n",i,*(a+i));
	}
	return 0;

}
```

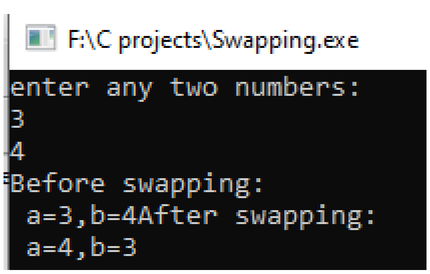
Output:

D:\Lab Report c From Structure\file in c\pointer and array.exe

```
how many elements do you wanna store:
5
enter elements for the array:
4
5
35
45
33
array[0]=4
array[1]=5
array[2]=35
array[3]=45
array[4]=33
```

```
//Swapping using pass by reference.
#include<stdio.h>
void swap(int *,int *);
int main(){
	int a,b;
	printf("enter any two numbers: \n");
	scanf("%d%d",&a,&b);

	swap(&a,&b);
	printf("After swapping:\n a=%d,b=%d",a,b);
	return 0;
}
void swap(int *a,int *b){
	printf("Before swapping:\n a=%d,b=%d",*a,*b);
	int temp;
	temp=*a;
	*a=*b;
	*b=temp;
}
```

Output:



```c
//pointer to pointer (chain of pointer)
#include<stdio.h>
int main(){
	int n;
	printf("enter any number\n");
	scanf("%d",&n);
	int *p;
	p=&n;
	int **q;
	q=&p;
	int ***d;
	d=&q;
	printf("address of pointer 1 is %d\n",p);
	printf("address of chain of pointer is :%d\n",q);
	printf("address of chain of pointer is :%d\n",d);
	printf("value using pointer 1: %d\n",*p);
	printf("value using chain of pointer: %d\n",**q);
	printf("value using chain of pointer: %d\n",***d);
}
```

Output:

F:\C projects\chain of pointer.exe

```
enter any number
3
address of pointer 1 is 6487572
address of chain of pointer is :6487560
address of chain of pointer is :6487552
value using pointer 1: 3
value using chain of pointer: 3
value using chain of pointer: 3
```

```
//without using DMA;
#include <stdio.h>

struct pointer {
   int age;
   char name[30];
};
int main() {
   struct pointer p1;
   struct pointer *p;
   p = &p1;
   printf("Enter your name: ");
   scanf(" %[^\n]", p->name);
   printf("Enter age: ");
   scanf("%d", &p->age);
   printf("\nAge is: %d\n", p->age);
   printf("Name is: %s\n", p->name);

   return 0;
}
```

Output:

F:\C projects\without using DMA.exe

```
Enter your name: helleodf
Enter age: 45

Age is: 45
Name is: helleodf
```

```c
//Demo to DMA and Malloc .
#include <stdio.h>
#include <stdlib.h>

struct pointer {
   int age;
   char name[30];
} *p;

int main() {
   p = (struct pointer *)malloc(sizeof(struct pointer));
           if (p == NULL) {
      printf("Memory allocation failed!\n");
    return 1;
   }
   printf("Enter your name: ");
   scanf(" %[^\n]s", p->name);
           printf("Enter age: ");
   scanf("%d", &p->age);
   printf("\nAge is: %d\n", p->age);
   printf("Name is: %s\n", p->name);
           free(p);
   return 0;
}
```

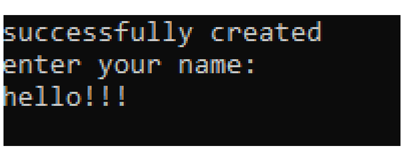
Output:

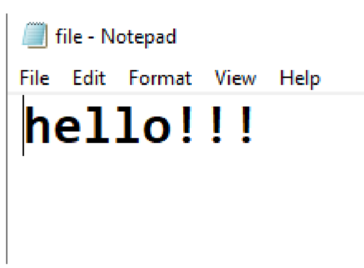
F:\C projects\without using DMA.exe

```
Enter your name: helloo
Enter age: 43

Age is: 43
Name is: helloo
```

```
//creating and storing information in a file
#include<stdio.h>
#include<stdlib.h>

int main(){
	FILE *f;
	f= fopen("file.txt","w");
	if(f==NULL){
		printf("failed to create \n");
	}
	else{
		printf("successfully created\n");
	}
	char name[30];
	printf("enter your name:\n");
	scanf("%[^\n]s",name);
	fprintf(f,"%s",name);
return 0;
}
```

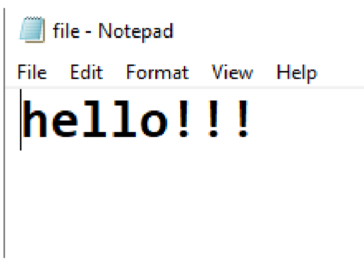
Output:

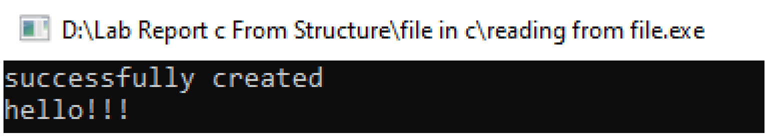




```
//reading the existing content of file
#include<stdio.h>
#include<stdlib.h>

int main(){
	FILE *f;
	f= fopen("file.txt","r");
	if(f==NULL){
		printf("failed to create \n");
	}
	else{
		printf("successfully created\n");
	}
	char name[30];
	fscanf(f,"%s",name);
	printf("%s",name);
return 0;
}
```

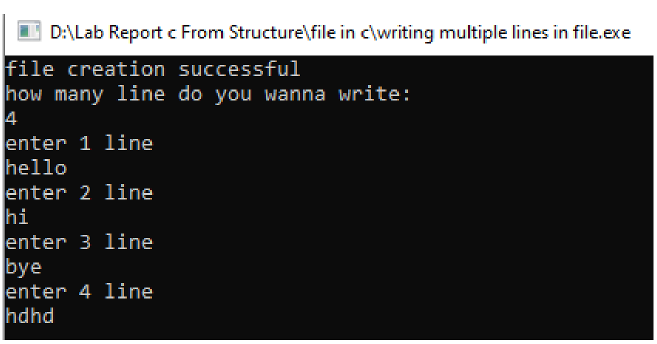
<u>Output:</u>

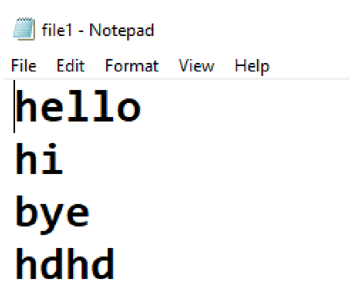




```
//writing multiple lines in file

#include<stdio.h>
#include<stdlib.h>

int main(){
	FILE *f;
	f=fopen("file1.txt","w");
	if(f==NULL){
		printf("file creation unsuccessful\n");
	}
	else{
		printf("file creation successful\n");
	}
	int n;
	printf("how many line do you wanna write:\n");
	scanf("%d",&n);
	char str[50];
	int i;
	for(i=1;i<=n;i++){
		printf("enter %d line \n",i);
		scanf("%s",str);
		fprintf(f,"%s\n",str);
	}
	return 0;
}
```

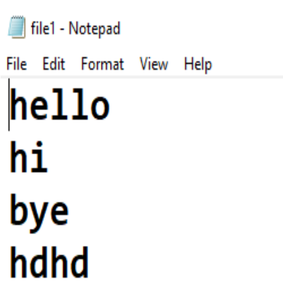
Output:

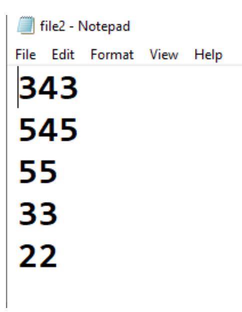


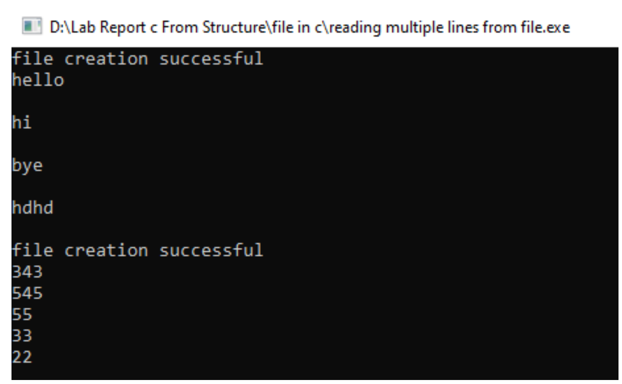


```
//reading multiple lines from file
#include<stdio.h>
#include<stdlib.h>
int main(){
	FILE *f;
	f=fopen("file1.txt","r");
	if(f==NULL){
		printf("file creation unsuccessful\n");
	}
	else{
		printf("file creation successful\n");
	}
	char name[50];
	while(fgets(name,50,f)!=0){
		printf("%s\n",name);
	}
	fclose(f);
	FILE *p;
		p=fopen("file2.txt","r");
	if(p==NULL){
		printf("file creation unsuccessful\n");
	}
	else{
		printf("file creation successful\n");
	}
	int n;
	while(fscanf(p,"%d",&n)==1){
		printf("%d\n",n);
	}
	fclose(p);
	return 0;
}
```

Output:







//WAP to input 10 numbers from user and store it on a file named odd.txt and read that file and print only odd number

```
#include<stdio.h>
#include<stdlib.h>
int main(){
	FILE *f;
	f = fopen("odd.txt","w");
	if(f==NULL){
		printf("failed to create file\n");
	}
	else{
		printf("successfully opened file\n");
	}
	int n[10],i;
	printf("enter any 10 numbers\n");
	for(i=0;i<10;i++){
		scanf("%d",&n[i]);
		fprintf(f,"%d\n",n[i]);
	}
	fclose(f);
	FILE *p;
	f = fopen("odd.txt","r");
	p = fopen("write.txt","w");
	if(p==NULL){
		printf("failed to create file\n");
	}
	else{
		printf("successfully opened file\n");
	}
	int a;
	while(fscanf(f,"%d",&a)==1){
		if(a%2!=0){
			printf("odd value : %d\n",a);
			fprintf(p,"%d\n",a);
		}
	}
	fclose(f);
	return 0;
}
```

Output:

D:\Lab Report c From Structure\file in c\demo.exe

```
successfully opened file
enter any 10 numbers
1
2
3
4
5
6
7
8
9
13
successfully opened file
odd value : 1
odd value : 3
odd value : 5
odd value : 7
odd value : 9
odd value : 13
```

write - Notepad

File Edit Format View Help

```
1
3
5
7
9
13
```

odd - Notepad

File Edit Format View Help

```
1
2
3
4
5
6
7
8
9
13
```